

Dong Han

CONTACT INFORMATION

Düsseldorf,
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WORKING EXPERIENCE

PhD Candidate - HUAWEI - Germany

3/2023–Now

- Conducting cutting-edge projects and research in trustworthy AI, privacy-preserving and data security.
- Investigating the applications of trustworthy AI in computer vision tasks.
- Publishing research papers in international conferences.

Master Thesis Research Intern - Laboratoire Hubert Curien - France

1/2022–8/2022

- The objective of the project is to investigate the illumination spectral power distribution (SPD) estimation.
- The modified image signal processing (ISP) pipeline is implemented for modeling the sensor imaging.
- The synthetic dataset and captured dataset are proposed as SPD training database.
- The deep learning model 1D CNNs is proposed for illumination estimation in order to predict lighting conditions in an unknown scenario.

3D Printing Research Intern - Mihaly - France

6/2021–7/2021

- The goal was to achieve certain similar glossiness in terms of human skin by using OpenCV to implement the algorithm and CMYK printer for printing.
- 3D skin printing can be a potential application since human skin has a certain microstructure that affects the most perception of customers.
- **Output:** The developed algorithm was able to simulate the glossiness of images, which was used for further 3D printing.

EDUCATION

Friedrich Schiller University Jena, Germany

2023–Now

- PhD in Privacy-Preserving Computer Vision.
- Funded by company.
- Thesis supervisor: Prof. Joachim Denzler.

Norwegian University of Science and Technology - Norway

2020–2022

University of Granada - Spain

University of Eastern Finland - Finland

Jean Monnet University - France

- M.Sc. in Computational Colour and Spectral Imaging (COSI)
- Erasmus Mundus Joint Master Degrees Program.
- Full scholarship (49000 euros).
- Relevant courses: Computer Vision, Advanced Color and Image Processing, Data Science, Deep Learning for Visual Computing, Human perception and cognition, Advanced Color and Spectral Imaging, Color Science, Advanced Deep Learning.
- Master thesis: Deep Learning-Based Illumination SPD Estimation from an RGB Camera
- Average grade: B

Dalian University, Dalian

2016–2020

- B.Sc. in Optoelectronic Information Science and Engineering, full-time program.
- Overall GPA: 87/100.
- Bachelor thesis: Research on Optical Properties of Bimetal Nanospheres.

- **High-Quality Face Image Restoration via Diffusion Models** 11/2024–07/2025
Huawei (Germany)
 - Aimed to reconstruct complete, high-fidelity facial images from facial feature embeddings (vectors).
 - Proposed a vector mapping method based on Kolmogorov-Arnold Networks (KAN) to significantly enhance the efficiency of facial generation models.
 - Achieved restoration performance exceeding current State-of-the-Art (SOTA) models; the framework serves as a validation tool for evaluating the security of arbitrary face recognition systems.
- **Content Safety in Text-to-Image Models via Reinforcement Learning and Machine Unlearning** 05/2024–07/2025
Huawei (Germany)
 - Focused on mitigating and controlling the generation of harmful or NSFW content in text-to-image (T2I) models.
 - Proposed a Reinforcement Learning-based LoRA fine-tuning technique combined with partial Self-Attention layer mapping to erase prohibited concepts.
 - Outperformed SOTA models in harmful content reduction rate, with high generalizability to Image-to-Image (I2I) tasks.
- **Facial Privacy Protection Based on Diffusion Models** 02/2024–11/2024
Huawei (Germany)
 - Addressed privacy risks associated with sharing facial images on social media platforms.
 - Developed a training-free privacy protection mechanism leveraging Diffusion-based Image-to-Image (I2I) generation and guidance scale adjustment.
 - Surpassed SOTA protection benchmarks while maintaining image naturalness without requiring additional model training.
- **Privacy-Preserving Face Recognition Integrating Frequency and Color Domains** 03/2023–11/2023
Huawei (Germany)
 - Focused on protecting sensitive facial data using frequency domain information fusion and Differential Privacy (DP).
 - Proposed a fusion method for frequency and color information to further reduce channel dimensionality and computational overhead.
 - Balanced robust privacy protection with high face recognition accuracy, outperforming existing SOTA privacy-preserving recognition frameworks.
- **Semantic Segmentation on Spectral Image with Deep Learning** 10/2021–11/2021
University of Eastern Finland (Finland)
 - The goal was to do segmentation on human placenta tissue hyperspectral images by using deep learning methods.
 - The variety of machine learning dimensionality reduction (PCA, MNF, ICA) was done as a data pre-processing step. The different deep learning models (UNet, UNet 3+, Swin-Unet) were implemented and compared.
 - The GUI was designed based on PySide2 for spectral image visualization and other basic image processing modules were integrated.
- **Spectral Image Segmentation and Human Perception** 5/2021–6/2021
University of Granada (Spain)
 - The goal was to do segmentation on spectral images and design a psychophysical experiment using corresponding RGB images by the Psychtoolbox Framework.
 - The segmentation was conducted on the spectral image by using K-means clustering algorithms and the Jaccard index was used to evaluate the results.
 - The psychophysical experiment was designed to study the effect of hue shifting and chroma shifting on human perceptions.
- **Dilated CNN in Variable Splitting Network for Medical Image Denoising** 9/2020–12/2020
Norwegian University of Science and Technology (Norway)
 - The goal of this project was to implement and enhance the denoising process in image reconstruction.
 - The CS-MRI concept was implemented for medical image reconstruction.
 - A dilated CNN was applied for improving denoising performance.

PUBLICATION

- **Realistic Face Reconstruction from Facial Embeddings via Diffusion Models**
 - AAAI, 2026.
- **Diffusion-based Identity-Preserving Facial Privacy Protection**
 - **Han, Dong**, et al. "Diffusion-based Identity-Preserving Facial Privacy Protection." ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**). IEEE, 2025.
- **Robust Skin Color Driven Privacy Preserving Face Recognition via Function Secret Sharing**
 - **Han, Dong**, et al. "Robust skin color driven privacy-preserving face recognition via function secret sharing." 2024 IEEE International Conference on Image Processing (**ICIP**). IEEE, 2024..
- **Privacy-Preserving Face Recognition in Hybrid Frequency-Color Domain** 27/2/2024
 - **Han, D.**; Li, Y. and Denzler, J. (2024). Privacy-Preserving Face Recognition in Hybrid Frequency-Color Domain. In Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications.
- **Lighting Spectral Power Distribution Estimation With RGB Camera** 26/2/2023
 - **Han, Dong**, et al. "Lighting Spectral Power Distribution Estimation with RGB Camera." 2022 16th International IEEE Conference on Signal-Image Technologies and Internet-Based System. 2022.
- **Medical Image Denoising in MRI Reconstruction Procedure** 11/9/2021
 - **Han, Dong**, and Ronny Velastegui. "Medical Image Denoising in MRI Reconstruction Procedure." International Conference on Computational Science and Its Applications. Springer, Cham, 2021.
- **The Importance of Color Spaces for Image Classification Using Artificial Neural Networks: A Review** 11/9/2021
 - Velastegui, Ronny, Linna Yang, and **Dong Han**. "The Importance of Color Spaces for Image Classification Using Artificial Neural Networks: A Review." International Conference on Computational Science and Its Applications. Springer, Cham, 2021.

TECHNICAL SKILLS

- Programming Languages: Python, R, C++, LaTeX.
- Technical Softwares/Frameworks: PyTorch, OpenCV, Tensorflow, Keras, Scikit-Learn, PySide2, Git, MATLAB, Visual Studio Code.

LANGUAGE

- Languages: Chinese (native speaker), English (IELTS 6.5), German (A2).

AWARDS

- Huawei Rising Star (2025)
- Erasmus Mundus Master Scholarship (60 students selected in China) (2020)
- First Prize of Photoelectric Design Competition (Object recognition from speckle image) (2019)
- WangTeng Scholarship of Physics Faculty (for outstanding performance in math and physics) (2018)
- Best Student (top 8% at faculty) (2018)
- Second Prize of Computer Design Competition in Liaoning Province (2017)
- Third Prize of The Ninth National College Student Mathematics Competition (2017)